# IPL Data Analysis (2008–2024)

This project analyzes historical Indian Premier League (IPL) data from 2008 to 2024. We explore player performances, team statistics, toss impact, and venue insights using Python.

**Tools Used**: Python, Pandas, Matplotlib, Seaborn
**Datasets**: `matches.csv`, `deliveries.csv`

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set plotting style
sns.set(style = 'whitegrid')

# Load datasets
matches = pd.read_csv('E:\matches.csv')
deliveries = pd.read_csv('E:\deliveries.csv')

# Quick preview
matches.head(), deliveries.head()
```

```
(       id   season        city        date match_type player_of_match  \
 0  335982  2007/08   Bangalore  2008-04-18     League     BB McCullum
 1  335983  2007/08  Chandigarh  2008-04-19     League      MEK Hussey
 2  335984  2007/08       Delhi  2008-04-19     League     MF Maharoof
 3  335985  2007/08      Mumbai  2008-04-20     League      MV Boucher
 4  335986  2007/08     Kolkata  2008-04-20     League       DJ Hussey

                                     venue                       team1  \
 0                     M Chinnaswamy Stadium  Royal Challengers Bangalore
 1  Punjab Cricket Association Stadium, Mohali              Kings XI Punjab
 2                           Feroz Shah Kotla             Delhi Daredevils
 3                           Wankhede Stadium               Mumbai Indians
 4                               Eden Gardens        Kolkata Knight Riders

                        team2                  toss_winner toss_decision  \
 0        Kolkata Knight Riders  Royal Challengers Bangalore         field
 1          Chennai Super Kings          Chennai Super Kings           bat
 2            Rajasthan Royals             Rajasthan Royals           bat
 3  Royal Challengers Bangalore               Mumbai Indians           bat
 4             Deccan Chargers              Deccan Chargers           bat

                       winner   result  result_margin  target_runs  \
 0        Kolkata Knight Riders     runs          140.0        223.0
 1          Chennai Super Kings     runs           33.0        241.0
 2             Delhi Daredevils  wickets            9.0        130.0
 3  Royal Challengers Bangalore  wickets            5.0        166.0
 4        Kolkata Knight Riders  wickets            5.0        111.0

    target_overs super_over method    umpire1         umpire2
 0          20.0          N    NaN  Asad Rauf     RE Koertzen
 1          20.0          N    NaN  MR Benson      SL Shastri
 2          20.0          N    NaN  Aleem Dar  GA Pratapkumar
 3          20.0          N    NaN   SJ Davis       DJ Harper
 4          20.0          N    NaN  BF Bowden     K Hariharan  ,
    match_id  inning           batting_team                 bowling_team  over  \
 0    335982       1  Kolkata Knight Riders  Royal Challengers Bangalore     0
 1    335982       1  Kolkata Knight Riders  Royal Challengers Bangalore     0
 2    335982       1  Kolkata Knight Riders  Royal Challengers Bangalore     0
 3    335982       1  Kolkata Knight Riders  Royal Challengers Bangalore     0
 4    335982       1  Kolkata Knight Riders  Royal Challengers Bangalore     0

    ball       batter   bowler  non_striker  batsman_runs  extra_runs  \
 0     1   SC Ganguly  P Kumar  BB McCullum             0           1
 1     2  BB McCullum  P Kumar   SC Ganguly             0           0
 2     3  BB McCullum  P Kumar   SC Ganguly             0           1
 3     4  BB McCullum  P Kumar   SC Ganguly             0           0
 4     5  BB McCullum  P Kumar   SC Ganguly             0           0

    total_runs extras_type  is_wicket player_dismissed dismissal_kind fielder
 0           1     legbyes          0              NaN            NaN     NaN
 1           0         NaN          0              NaN            NaN     NaN
 2           1       wides          0              NaN            NaN     NaN
 3           0         NaN          0              NaN            NaN     NaN
 4           0         NaN          0              NaN            NaN     NaN  )
```

```
# Check missing values
print("Matches:\n", matches.isnull().sum())
print("Deliveries:\n", deliveries.isnull().sum())

# Drop unneeded columns
#matches.drop(['method','umpire1','umpire2'], axis = 1, inplace = True)

# Replace team names
team_mapping = {
    'Mumbai Indians': 'MI', 'Kolkata Knight Riders': 'KKR',
    'Royal Challengers Bangalore': 'RCB', 'Royal Challengers Bengaluru': 'RCB',
    'Deccan Chargers': 'DCR','Chennai Super Kings': 'CSK',
    'Rajasthan Royals': 'RR', 'Sunrisers Hyderabad': 'SRH',
    'Delhi Daredevils': 'DD', 'Delhi Capitals': 'DC',
    'Gujarat Lions': 'GL', 'Gujarat Titans': 'GT',
    'Kings XI Punjab': 'KXIP', 'Punjab Kings': 'PBK',
    'Rising Pune Supergiants': 'RPS', 'Rising Pune Supergiant': 'RPS',
    'Kochi Tuskers Kerala': 'KTK', 'Lucknow Super Giants' : 'LSG',
    'Pune Warriors': 'PW', 'Delhi Capitals': 'DC'
}
matches.replace(team_mapping, inplace = True)
deliveries.replace(team_mapping, inplace = True)
```

```
Matches:
 id                    0
season                0
city                 51
date                  0
match_type            0
player_of_match       5
venue                 0
team1                 0
team2                 0
toss_winner           0
toss_decision         0
winner                5
result                0
result_margin        19
target_runs           3
target_overs          3
super_over            0
method             1074
umpire1               0
umpire2               0
dtype: int64
Deliveries:
 match_id                 0
inning                   0
batting_team             0
bowling_team             0
over                     0
ball                     0
batter                   0
bowler                   0
non_striker              0
batsman_runs             0
extra_runs               0
total_runs               0
extras_type         246795
is_wicket                0
player_dismissed    247970
dismissal_kind      247970
fielder             251566
dtype: int64
```

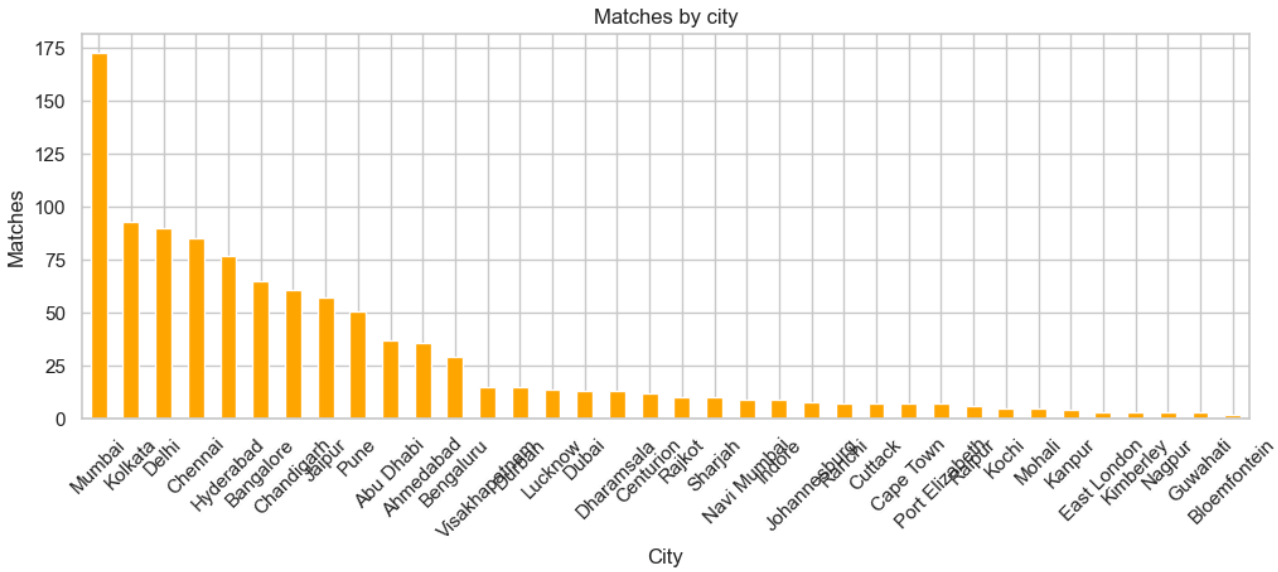
## Exploratory Data Analysis

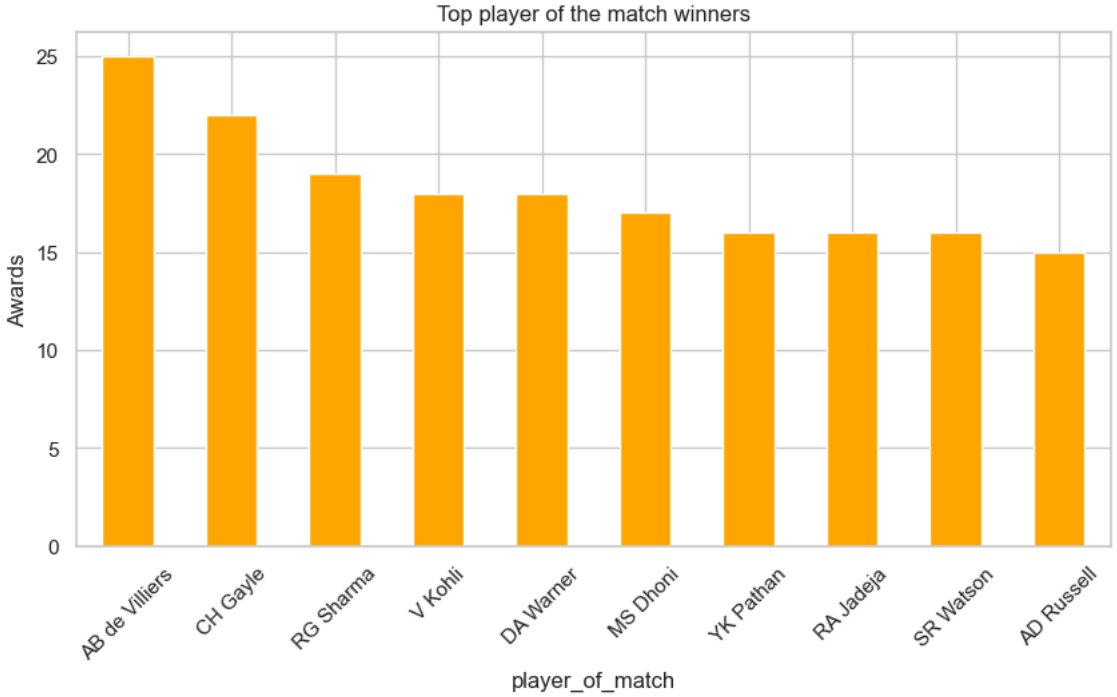
- Match insights

```
# Total matches and venues
print("Total matches:", matches.shape[0])
print("Total venues:", matches['venue'].nunique())

# Matches per city
matches['city'].value_counts().plot(kind = 'bar', color = 'orange', figsize = (10, 5), title = 'Matches by city')
```

```
plt.tight_layout()
plt.xlabel('City')
plt.ylabel('Matches')
plt.xticks(rotation = 45)
plt.show()

# Player of the match leaders
matches['player_of_match'].value_counts().head(10).plot(kind = 'bar', color = 'orange', figsize = (10, 5))
plt.title('Top player of the match winners')
plt.ylabel('Awards')
plt.xticks(rotation = 45)
plt.show()
```

```
Total matches: 1095
Total venues: 58
```

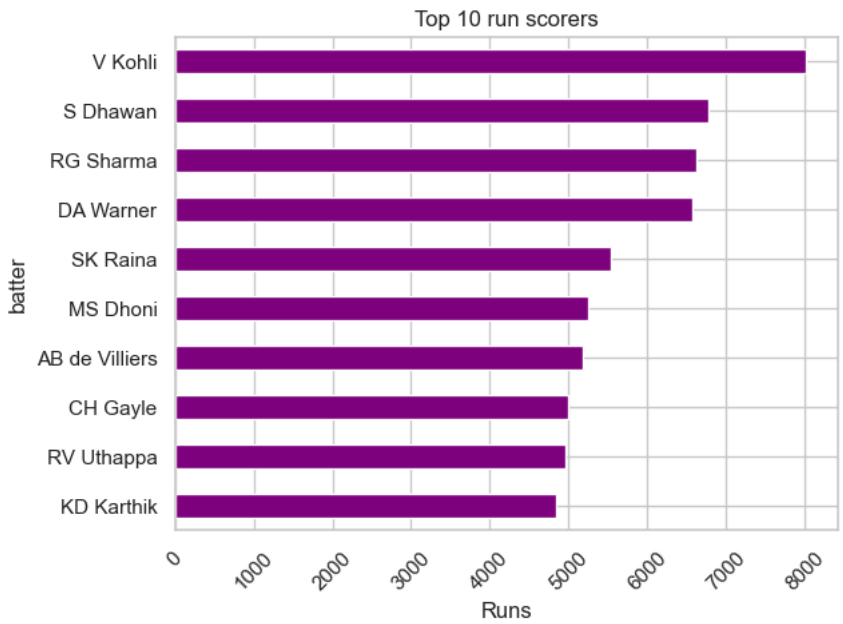


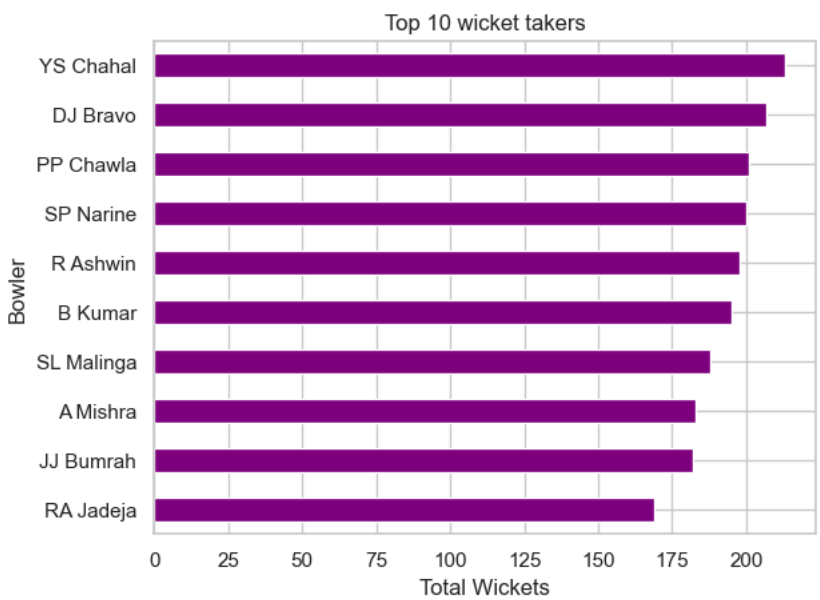




- Batting and bowling stats

```
# Top run scorers
top_run_scorers = deliveries.groupby('batter')['batsman_runs'].sum().sort_values(ascending = False).head(10)
top_run_scorers.plot(kind = 'barh', color = 'purple')
plt.title('Top 10 run scorers')
plt.xlabel('Runs')
```

```
plt.xticks(rotation = 45)
plt.gca().invert_yaxis()
plt.show()

# Top wicket takers
wickets = deliveries[deliveries['dismissal_kind'].notnull()]
top_wicket_takers = wickets.groupby('bowler')['player_dismissed'].count().sort_values(ascending = False).head(10)
top_wicket_takers.plot(kind = 'barh', color = 'purple')
plt.title('Top 10 wicket takers')
plt.xlabel('Total Wickets')
plt.ylabel('Bowler')
plt.gca().invert_yaxis()
plt.show()
```




- Toss analysis

```
# Toss winner also winning the match
toss_win_match_win = matches[matches['toss_winner'] == matches['winner']]
percentage = len(toss_win_match_win) / len(matches) * 100
print(f'Toss winner who also win the match: {percentage:.2f}%')
```

```
Toss winner who also win the match: 50.59%
```

- Consistant performers across season

```
# Total runs per batsman per season
runs_per_season = deliveries.merge(matches[['id','season']], left_on = 'match_id', right_on = 'id').groupby(['season','batter']

# Top 5 consistent batsman (appearing in top 10 across most seasons)
top_performers = runs_per_season.groupby('season', group_keys = False).apply(lambda x: x.nlargest(10, 'batsman_runs'), include_

# Count appearence in top 10
consistant_batsman = top_performers['batter'].value_counts().head(5)

print("Runs per season\n", runs_per_season, "\n")
print("Top performers\n", top_performers, "\n")
print("Consitant batsman\n", consistant_batsman, "\n")
```

```
Runs per season
        season             batter  batsman_runs
0     2007/08           A Chopra            42
1     2007/08           A Kumble            13
2     2007/08           A Mishra            37
3     2007/08           A Mukund             0
4     2007/08            A Nehra             3
...       ...                ...           ...
2612     2024           WG Jacks           230
2613     2024            WP Saha           136
2614     2024  Washington Sundar             0
2615     2024        YBK Jaiswal           435
2616     2024      Yudhvir Singh            21

[2617 rows x 3 columns]

Top performers
                batter  batsman_runs
0           SE Marsh           616
1          G Gambhir           534
2      ST Jayasuriya           514
3          SR Watson           472
4           GC Smith           441
..               ...           ...
165  B Sai Sudharsan           527
166         KL Rahul           520
167         N Pooran           499
168        SP Narine           488
169  Abhishek Sharma           484

[170 rows x 2 columns]

Consitant batsman
 batter
SK Raina     9
V Kohli      9
S Dhawan     8
DA Warner    7
RG Sharma    6
Name: count, dtype: int64
```

- Death over specialists

```
# Filter to overs 16-20 (death overs)
death_overs = deliveries[deliveries['over'] > 16]

# Most wickets in death overs
death_wickets = death_overs[death_overs['dismissal_kind'].notnull()].groupby('bowler').size().sort_values(ascending = False).he
death_wickets
```

```
bowler
DJ Bravo          103
SL Malinga         90
B Kumar            80
HV Patel           68
JJ Bumrah          68
MM Sharma          63
Mohammed Shami     57
SP Narine          54
R Vinay Kumar      50
CH Morris          50
dtype: int64
```

## Key Insights

- **Venue Trends**: Mumbai, kolkata and delhi are top match-hosting cities.
- **Top Players**: Consistent performers like V Kohli and S Dhavan are frequent match-winners.
- **Toss Impact**: Toss winners win 50.59% of the time; toss isn't always a game-decider.
- **Wicket Kings**: Bowlers like chahal and Bravo dominate wickets across seasons.
- **Seasonal Consistency** : Virat Kohli and Suresh Raina frequently rank among top 10 run-scorers—highlighting their sustained performance across multiple IPL seasons.
- **Death Over Dominance** : Lasith Malinga and Dwayne Bravo are top wicket-takers in overs 16–20, proving their effectiveness under late- innings pressure.

## Conclusion

This analysis reveals patterns in IPL match outcomes, team strategies, and individual performances. It offers useful insights for cricket analysts and fans alike.

### Future Improvements:

- Add interactive dashboards (Plotly/Streamlit)
- Predictive modeling: match winners or top players
- Deeper player-by-player breakdown